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Geopolitical Risk and Systemic Risk in Global Systemically Important Banks (G-SIBs)

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Abstract

This research studies how geopolitical risk relates to the systemic vulnerability of global systemically important banks (G-SIBs). Following the market-based SRISK framework of Brownlees and Engle (2017) and measuring geopolitical risk using the news-based GPR index of Caldara and Iacoviello (2022), the analysis documents the time-series co-movement between geopolitical risk and aggregate systemic capital shortfall, using a DCC-based model (Engle, 2002) to capture time-varying dependence. The results show that spikes in geopolitical risk sometimes coincide with higher aggregate systemic expected capital shortfall, but that this relationship is unstable over time. This research was conducted at the Financial Stability - Macroprudential Policy and Analysis department of De Nederlandsche Bank and provides context that supports systemic risk monitoring and informs the calibration and targeting of macroprudential policy tools.

Statement of Originality

This document is written by Student Thijs de Jong who declares to take full responsibility for the contents of this document. I declare that the text and the work presented in this document is original and that no sources other than those mentioned in the text and its references have been used in creating it. I have not used generative AI (such as ChatGPT) to generate or rewrite text. UvA Economics and Business is responsible solely for the supervision of completion of the work and submission, not for the contents.

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Chapter 1

Introduction

Geopolitical events can quickly shake financial markets. News about wars, rising military tensions, sanctions, or diplomatic conflicts often affect investors' expectations about future economic growth and financial conditions (Caldara & Iacoviello, 2022). When uncertainty rises, investors reassess risk and rebalance their portfolios, which can lead to sharp movements in asset prices and tighter financial conditions (Bloom, 2009; Pástor & Veronesi, 2013). Because these shocks can affect many assets and institutions at the same time, they matter not only for individual investors and firms, but also for the stability of the financial system as a whole.

The connection between geopolitical risk and financial stability is especially relevant for the banking sector. Their distress can spread quickly to the rest of the financial system, particularly when many institutions are hit at the same time (Acemoglu et al., 2015; Acharya et al., 2017). When this occurs, the broader costs to society can be much larger than the individual costs faced by the institutions, which motivates macroprudential monitoring and regulation (Acharya et al., 2017; Hanson et al., 2011).

Recent studies are increasingly pointing to a clear relationship between geopolitical risk and systemic risk in the financial sector. For example, Gabbiadini et al. (2025) show that geopolitical risk is associated with higher market-based measures of systemic risk in European banks. Other research finds similar patterns, suggesting that geopolitical events can weaken bank stability and influence lending conditions (Demir & Danisman, 2021; Nguyen & Thuy, 2023; Phan et al., 2022).

Evidence on how geopolitical risk is reflected in system-wide capital shortfalls remains limited, particularly for globally systemically important banks. This gap matters, especially given the growing policy attention on how geopolitical developments and financial stability interact (Chițu et al., 2022). There is also evidence that regulation plays a role in shaping how strongly geopolitical risk feeds into systemic risk, which makes the policy environment an important part

of the story (Wang et al., 2025).

This thesis contributes by linking geopolitical risk to a prudentially interpretable measure of systemic risk (SRISK) in a sample of global systemically important banks (G-SIBs). The central research question is:

How is geopolitical risk related to aggregate SRISK in the G-SIB sector, and how do the regional composition and concentration of SRISK evolve over time?

To address this question, the analysis proceeds in three steps. First, it documents time-series co-movement between geopolitical risk and aggregate SRISK. Next, it tests whether geopolitical risk has incremental explanatory power for changes in aggregated SRISK after accounting for persistence. Third, it explores how the distribution of aggregate SRISK evolves over time by decomposing SRISK across regions and analyzing concentration among the largest contributors.

The decision to focus on G-SIBs is driven by their importance for global financial stability and by the policy relevance of monitoring systemic vulnerability within this group of banks. Each year, the Financial Stability Board (FSB) identifies G-SIBs using indicators such as size, interconnectedness, complexity, and cross-border activity, using the methodology developed by the Basel Committee (Basel Committee on Banking Supervision, 2024; Financial Stability Board, 2025). Studying G-SIBs therefore provides a direct link between market-based systemic risk measures and macroprudential concerns.

Systemic vulnerability is measured using SRISK, which captures the expected capital short-fall of a bank in a severe market decline and can be aggregated across institutions (Brownlees & Engle, 2017). The empirical analysis combines the GPR index with market and balance-sheet data for the 2025 G-SIB sample (Financial Stability Board, 2025). The sample covers the period from 2006 to 2025, and the market returns are proxied by the MSCI World Banks index (MSCI Inc., n.d.). The goal is descriptive and monitoring-oriented, providing context for macroprudential policy when geopolitical tensions are elevated.

The remainder of this thesis is organized as follows. Chapter 2 reviews the key literature on geopolitical risk, uncertainty shocks, and how systemic risk is measured. Chapter 3 presents the empirical framework, including the DCC-GARCH model and the SRISK methodology. Chapter 4 introduces the main data sources: the GPR index, the G-SIB sample, and the market index used throughout the analysis. Chapter 5 presents the empirical results on aggregate co-movement and predictability after accounting for persistence. It also examines how SRISK evolves in terms of regional contributions and concentration among the largest banks. Finally, Chapter 6 summarizes the main findings and discusses implications for monitoring systemic risk in a world where geopolitical tensions are increasingly in the spotlight.

Chapter 2

Literature Review

2.1 Geopolitical Risk

Geopolitical risk refers to the threat, realization, and escalation of adverse international events such as wars, military standoffs, terrorism, sanctions, and major diplomatic crises (Caldara & Iacoviello, 2022). These events can quickly change expectations about growth, inflation, and financial stability (Caldara & Iacoviello, 2022). Because major geopolitical episodes are hard to capture with simple event dummies, recent work increasingly relies on text-based measures that turn news coverage into a continuous index, such as the GPR index (Caldara & Iacoviello, 2022) and the EPU index (Baker et al., 2016). An advantage of the GPR index is that it captures not only the intensity of geopolitical tensions but also whether news coverage is dominated by threats or by realized events (Caldara & Iacoviello, 2022). This matters for financial stability because geopolitical risk is multifaceted and its different dimensions can affect banks through different, and potentially reinforcing, transmission channels (Catalán & Tsuruga, 2023). Consistent with this, empirical evidence suggests that higher geopolitical risk is associated with weaker bank stability (Phan et al., 2022).

Three main channels highlighted in the literature are as follows. First, when geopolitical tensions rise, uncertainty increases, and firms and households often delay investment and spending (Bloom, 2009). This can increase banks' credit losses and make it more difficult for borrowers to repay their debts, which aligns with findings of Demir and Danisman (2021) that geopolitical risk affects bank credit. Tensions can also lead to trade restrictions that reduce profits for firms with global exposure (Caldara et al., 2020). Lastly, investors may quickly shift to safer assets in response to geopolitical shocks, which increases volatility and risk premiums and can tighten funding and liquidity conditions (Caldara & Iacoviello, 2022; Nguyen & Thuy, 2023). These effects tend to be larger when macro-financial conditions are already fragile (Brunner-

meier & Sannikov, 2014). Related work on political uncertainty in asset pricing suggests that such shocks can affect not only risk levels but also how strongly assets move together (Pástor & Veronesi, 2013). This matters for our analysis because systemic risk is about joint distress. Even moderate shocks can become important at the system level when many institutions are hit at the same time (Acemoglu et al., 2015), which calls for an approach that allows dependence to change over time.

Banks are a natural focus in this setting. Because banks are highly leveraged and rely on short-term funding, they can be sensitive to sudden changes in market prices and funding conditions (Brunnermeier & Pedersen, 2009). As Allen and Gale (2000) emphasize, modern banking systems are closely connected through interbank lending, overlapping portfolios and other linkages. Losses can spread when institutions sell similar assets at the same time or when funding becomes more expensive. Geopolitical shocks can therefore affect banks both directly, through asset prices, funding markets, and cross-border exposures, and indirectly, through their impact on firms and households (Caldara & Iacoviello, 2022; Catalán & Tsuruga, 2023).

These channels suggest that the effect of geopolitical risk on banks is time-varying. In calm periods markets may absorb shocks with limited spillovers, whereas in stressed periods the same shocks can lead to broad risk reduction and increase system-wide vulnerability (Longin & Solnik, 2001). Because volatility also increases in crises, changes in correlations should be interpreted carefully (Forbes & Rigobon, 2002). While existing work shows that geopolitical risk moves financial conditions and asset prices, there is less evidence on whether these shocks are reflected in systemic banking risk at short horizons. This thesis addresses this gap by testing whether geopolitical risk predicts expected systemic capital shortfalls for large banks. For banks, these geopolitical shocks can also act as a common factor that increases joint downside exposure to the market. This motivates the need for a clear framework to quantify systemic banking risk. The next section therefore reviews the literature on systemic risk in banking and the main approaches used to measure it.

2.2 Systemic Risk Measures

Following the global financial crisis (GFC), Acharya et al. (2017) argue that existing financial regulations such as Basel I and Basel II¹ are not sufficiently focused on systemic risk but rather address each institution's risk in isolation, even though their stated aim is to promote system-wide financial stability. Consistent with this critique, Acharya et al. (2012) emphasize

¹Basel I (1988) and Basel II (2004) are international regulatory frameworks developed by the Basel Committee on Banking Supervision, serving as primary standards for bank capital adequacy and stability.

that systemic risk should not be defined by the failure of a single institution but rather by its overall contribution to system-wide failure. While the failure of a single institution can often be absorbed by the system, simultaneous capital shortfalls across many institutions increase the risk to the economy as a whole.

These perspectives helped shift academic literature from balance-sheet-based indicators to dynamic, market-based measures for systemic risk (Bisias et al., 2012). In the context of this thesis, this shift is particularly relevant because major geopolitical events are difficult to anticipate and can trigger rapid spikes in market volatility (Caldara & Iacoviello, 2022). From a macroprudential point of view, what ultimately matters is whether banks have sufficient buffers to absorb such stress without reducing intermediation (Hanson et al., 2011). This motivates a measure that (i) is available at sufficiently high frequency to capture news-driven shocks, (ii) has a system-wide interpretation, and (iii) connects naturally to prudential concepts such as capital adequacy.

An early attempt to align systemic risk measurement with prudential concerns is the Systemic Expected Shortfall (SES; see Appendix Eq. 1) framework of Acharya et al. (2017). Formally, SES is the expected amount by which a financial institution's equity falls short of its "required" level conditional on a systemic event. SES links systemic vulnerability to leverage and to tail exposure. The latter is captured by the Marginal Expected Shortfall (MES), defined as the expected equity loss of a financial institution conditional on the market being in its left tail (Idier et al., 2014).

Empirically, the framework performs well at identifying vulnerable institutions ahead of stress episodes. Using pre-crisis data, Acharya et al. (2012) and Acharya et al. (2017) show that banks with a high SES (high MES and leverage) are those that later experience the largest equity losses and capital shortfalls during the GFC. Idier et al. (2014) finds that MES contains useful information about exposure to system-wide downturns, especially when combined with balance-sheet vulnerability.

A limitation is that SES depends on the calibration of the benchmark capital ratio and the definition of the systemic event, and it mainly captures exposure to aggregate tail risk rather than the extent to which distress propagates across institutions (Benoît et al., 2013). This motivates measures that model interconnectedness via conditional tail dependence, as well as capital-shortfall measures that translate tail exposure into an explicit shortfall using balance-sheet characteristics such as size and leverage.

An alternative approach to measuring systemic risk emphasizes conditional tail dependence across institutions. CoVaR (see Appendix Eq. 2), proposed by Adrian and Brunnermeier (2016),

measures the tail risk of a target entity j conditional on the state of entity i . It is typically estimated using quantile regression (Adrian & Brunnermeier, 2016; Koenker & Bassett Jr, 1978), which makes the approach straightforward to implement in high-frequency data. Empirically, CoVaR has been frequently used to identify systemically important institutions (Caliskan et al., 2021; Castro & Ferrari, 2014; Wu et al., 2023). However, CoVaR remains a return-based measure that does not map distress into an explicit capital shortfall and is therefore less useful for prudential policy. Moreover, standard implementations may miss non-linear relationships that often emerge during crises. These limitations motivate complementary measures that combine information on tail dependence with balance-sheet information such as capital and leverage.

Finally, the third measure considered is SRISK, introduced by Brownlees and Engle (2017). SRISK measures the expected capital shortfall of a financial institution during a severe market decline. In doing so, SRISK expresses systemic vulnerability in the amount of capital an institution would need to remain adequately capitalized in a crisis. Compared to SES and CoVaR, this is easy to interpret from a prudential perspective because it allows geopolitical risk to be studied in terms of changes in expected capital needs. SRISK is also additive across institutions (Brownlees & Engle, 2017), which makes it easy to build aggregate SRISK and decompose contributions in our sample of G-SIBs.

SRISK combines balance-sheet information with market-based tail risk. The latter is captured by LRMES, which is defined for a severe market decline over a fixed horizon (Brownlees & Engle, 2017). LRMES is typically estimated using models that allow risk and dependence to vary over time. Among other approaches, Brownlees and Engle (2017) use simulation methods based on the Dynamic Conditional Correlation (DCC) model of Engle (2002). This model captures time-varying correlations between institutions and the broader market, an essential feature, since correlations tend to spike during crises (Longin & Solnik, 2001) and amplify systemic risk (Naoui et al., 2010). As a result, the dynamic framework allows LRMES, and therefore SRISK, to remain responsive to shifting market conditions and to more accurately anticipate capital shortfalls.

Empirical evidence indicates that SRISK is informative in practice. Brownlees and Engle (2017) show that SRISK identifies institutions that were most vulnerable during the GFC and provides early warning signals of systemic distress. Related studies find that SRISK is related to crisis severity and performs well compared to SES and other market-based measures in several comparisons (Acharya et al., 2012; Engle et al., 2014), suggesting its usefulness for macroprudential supervision. While most of the empirical work on geopolitical risk focuses on returns or volatility (Caldara & Iacoviello, 2022), SRISK would allow us to research its relation

to bank capital shortfalls and system-wide vulnerability.

A drawback is that SRISK relies on equity-market information, which may be noisy during extreme periods (Brunnermeier & Pedersen, 2009). SRISK also inherits model risk from the GARCH–DCC framework used to estimate LRMES and may be sensitive to structural breaks (Pesaran et al., 2006). Nevertheless, SRISK best fits this thesis because it offers a high-frequency, prudentially interpretable measure of system-wide capital vulnerability, directly linking balance-sheet information and market-based tail risk to expected capital shortfalls and thus capturing the potential impact of geopolitical shocks.

Chapter 3

Model Description

This chapter outlines the empirical framework used in this research. Following the SRISK framework of Brownlees and Engle (2017), the analysis proceeds in three steps. First, a DCC-GARCH model is estimated on daily bank and market returns to obtain a time-varying conditional covariance structure. Second, this conditional covariance matrix is used to compute dynamic conditional betas, summarizing each bank’s exposure to broad market movements over time. Third, these time-varying risk dynamics are mapped into long-horizon tail exposure and ultimately into SRISK.

3.1 Dynamic Conditional Correlation Model

For market-based systemic risk measures such as SRISK, what matters is not only the level of bank volatility, but also how strongly bank equity co-moves with the market during periods of distress. Dependence tends to strengthen in extreme market conditions (Longin & Solnik, 2001), and correlation comparisons can be mechanically biased by heteroskedasticity (Forbes & Rigobon, 2002). This dependence is central to the SRISK framework of Brownlees and Engle (2017), where long-horizon tail exposure (LRMES) and the resulting capital shortfall increase when conditional volatilities and correlations rise.

To capture these features, we adopt the Dynamic Conditional Correlation Generalized Autoregressive Conditional Heteroskedasticity (DCC-GARCH) framework of Engle (2002). Relative to constant correlation (CCC) (Bollerslev, 1990) specifications, DCC allows the full conditional covariance matrix between bank returns and market returns to vary over time. Fully parameterized multivariate GARCH models, such as BEKK (Engle & Kroner, 1995), quickly become infeasible as the number of institutions grows (Bauwens et al., 2006). Rolling-window correlations can be noisy and sensitive to the chosen window length. DCC-GARCH is a practi-

cal compromise. It is flexible enough to capture crisis-time co-movements, but scalable enough for panels of many banks.

DCC-GARCH delivers a time-varying conditional covariance matrix $\mathbf{H}_t$ defined as follows. Let $r_{k,t}$ denote the percentage log return of asset $k \in \{1, \dots, N\}$ at time t . The N assets consist of one market index and $N - 1$ banks. Without loss of generality, we denote the market by m and banks by i . In the empirical implementation, the market series is placed in the first position of $\mathbf{r}_t$, but in notation we refer to it as m . Define the $N \times 1$ return vector

$$\mathbf{r}_t = \begin{bmatrix} r_{1,t} \\ \vdots \\ r_{N,t} \end{bmatrix} \quad (3.1)$$

where $r_{i,t} = 100 \cdot \log(P_{i,t}/P_{i,t-1})$. Prior to estimation, each return series is demeaned such that its sample mean equals zero. Let $\tilde{r}_{i,t} \equiv r_{i,t} - \bar{r}_i$ denote the demeaned return of asset i , and define the $N \times 1$ demeaned return vector $\tilde{\mathbf{r}}_t = (\tilde{r}_{1,t}, \dots, \tilde{r}_{N,t})'$. For notational convenience, we drop the tilde in what follows and write $\mathbf{r}_t$ for demeaned returns. We therefore model returns as

$$\mathbf{r}_t = \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{H}_t). \quad (3.2)$$

where the Gaussian specification is used as a working conditional density for QMLE¹ (Bollerslev & Wooldridge, 1992). $\mathcal{F}_{t-1}$ denotes the information set at time $t - 1$ and $\mathbf{H}_t$ is the time-varying conditional covariance matrix. We adopt the DCC decomposition (Engle, 2002):

$$\mathbf{H}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t, \quad (3.3)$$

where $\mathbf{D}_t = \text{diag}(\sqrt{h_{1,t}}, \dots, \sqrt{h_{N,t}})$ collects the conditional standard deviations and $\mathbf{R}_t$ is the $N \times N$ time-varying conditional correlation matrix. This decomposition separates volatility dynamics from correlation dynamics.

First, the conditional variances $h_{i,t}$ are modeled by univariate GARCH(1,1) processes:

$$h_{i,t} = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i h_{i,t-1}, \quad i = 1, \dots, N, \quad (3.4)$$

with the constraints $\omega_i > 0$, $\alpha_i \geq 0$, $\beta_i \geq 0$ and $\alpha_i + \beta_i < 1$.

Second, the correlation dynamics are modeled using a DCC(1,1) specification based on standardized residuals:

$$\mathbf{z}_t = \mathbf{D}_t^{-1} \boldsymbol{\varepsilon}_t. \quad (3.5)$$

¹QMLE maximizes a Gaussian likelihood even when the true distribution may be non-normal. Under standard regularity conditions parameter estimates remain consistent, and inference can be based on robust standard errors.

Define the intermediate matrix $\mathbf{Q}_t$ as

$$\mathbf{Q}_t = (1 - a - b)\bar{\mathbf{Q}} + a\mathbf{z}_{t-1}\mathbf{z}_{t-1}^\top + b\mathbf{Q}_{t-1}, \quad (3.6)$$

where $a \geq 0$, $b \geq 0$, and $a + b < 1$. The parameters a and b capture the sensitivity and persistence of correlation dynamics. $\bar{\mathbf{Q}}$ represents the unconditional second-moment matrix of the standardized residuals. It is given by

$$\bar{\mathbf{Q}} = \frac{1}{T} \sum_{t=1}^T \mathbf{z}_t \mathbf{z}_t^\top \quad (3.7)$$

Because $\mathbf{Q}_t$ does not generally have unit diagonal, it is normalized to obtain a proper correlation matrix $\mathbf{R}_t$ by standardizing with its diagonal. Define $\mathbf{D}_{Q,t} = \text{diag}(q_{11,t}, \dots, q_{NN,t})$. Then

$$\mathbf{R}_t = \mathbf{D}_{Q,t}^{-\frac{1}{2}} \mathbf{Q}_t \mathbf{D}_{Q,t}^{-\frac{1}{2}}. \quad (3.8)$$

Together, $\mathbf{D}_t$ and $\mathbf{R}_t$ yield the conditional covariance matrix $\mathbf{H}_t$ in (3.3), which forms the basis for estimating time-varying betas and systemic risk measures in this analysis.

The parameters $\{\omega_i, \alpha_i, \beta_i\}_{i=1}^N$ and (a, b) are estimated by two-step QMLE, using a Gaussian distribution as a working likelihood.² The fitted sequences $\mathbf{H}_t$ and $\mathbf{R}_t$ are retrieved from the estimated DCC model.

3.2 Dynamic Conditional Beta

Let $r_{m,t}$ denote the market return and let $r_{i,t}$ denote the return of bank $i \neq m$. Following the dynamic conditional beta concept of Engle (2016), it is defined as the ratio of conditional second moments:

$$\beta_{i,t} = \frac{\text{Cov}_t(r_{i,t}, r_{m,t})}{\text{Var}_t(r_{m,t})} = \frac{h_{im,t}}{h_{mm,t}} \quad (3.9)$$

where $h_{im,t}$ and $h_{mm,t}$ are elements of the conditional covariance matrix $\mathbf{H}_t$, corresponding to bank-market conditional covariance and the market conditional variance, respectively.

Using the DCC decomposition $\mathbf{H}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$, the conditional covariance between bank i and the market can be written as

$$h_{im,t} = \rho_{im,t} \sqrt{h_{i,t} h_{m,t}}, \quad (3.10)$$

where $\rho_{im,t}$ is the (i, m) 'th element of the conditional correlation matrix $\mathbf{R}_t$, and $h_{i,t}$ and $h_{m,t}$ denote the corresponding conditional variances from the univariate GARCH components of the

²Estimation is implemented in R using `rugarch` and `rmgarch`, called from Python via `rpym2`.

DCC model. Substituting into the beta definition yields

$$\beta_{i,t} = \rho_{im,t} \sqrt{\frac{h_{i,t}}{h_{m,t}}}. \quad (3.11)$$

This expression makes clear that time variation in $\beta_{i,t}$ reflects both changes in conditional correlations and changes in relative conditional volatilities.

Note that $\beta_{i,t}$ is not used directly as an input in our baseline SRISK calculations. It is a derived summary of the same DCC-implied second moments ($h_{im,t}$ and $h_{m,t}$). Therefore, we report results based on $\beta_{i,t}$ only as a robustness check, to provide a benchmark against the baseline measures constructed directly from $\mathbf{H}_t$ in Section 3.4.

3.3 SRISK

Having obtained the time-varying conditional covariance matrix $\mathbf{H}_t$, the final step is to map the estimated return dynamics into SRISK (Brownlees & Engle, 2017). SRISK captures the expected capital shortfall of a financial institution in the event of a severe market decline. Intuitively, it measures how much capital a financial institution would need to raise in a crisis to meet regulatory requirements and remain adequately capitalized. Economically, SRISK depends on three elements: the institution's size, its leverage, and its sensitivity to broad market declines.

Formally, it is defined as:

$$\text{SRISK}_{i,t} = \max\{0, k(L_{i,t} + E_{i,t}) - (1 - \text{LRMES}_{i,t})E_{i,t}\} \quad (3.12)$$

where $L_{i,t}$ and $E_{i,t}$ denote the liabilities and market value of equity of institution i at time t , k is the SRISK prudential capital ratio threshold, and $\text{LRMES}_{i,t}$ is the Long-Run Marginal Expected Shortfall. In our baseline analysis, we set $k = 0.08$, a commonly used benchmark in the literature (Engle et al., 2014). The expression inside the braces is the institution's capital shortfall under stress: required capital $k(L_{i,t} + E_{i,t})$ minus the expected post-crisis equity value $(1 - \text{LRMES}_{i,t})E_{i,t}$. The $\max\{0, \cdot\}$ operator truncates negative values at zero so that only institutions with a positive shortfall contribute to aggregate SRISK. For institution-level interpretation, the untruncated capital shortfall is also considered. In this scenario, negative values can be read as “capital buffers” relative to the stress scenario.

3.4 LRMES

The risk-exposure input to SRISK is the Long-Run Marginal Expected Shortfall (LRMES), which measures the expected cumulative equity loss of institution i over a crisis horizon, conditional on a severe market decline (Brownlees & Engle, 2017).

Let $R_{m,t}^{(T)}$ denote the decimal log return of the market index over the next T trading days and $R_{i,t}^{(T)}$ the corresponding decimal log return of bank i . As daily returns in the DCC model are percent log returns r_t , the T -day decimal log returns satisfy $R_{j,t}^{(T)} = \sum_{s=1}^T r_{j,t+s}/100$ for $j \in \{m, i\}$. Following a benchmark discussed by Brownlees and Engle (2017), we set the crisis horizon to $T = 125$ trading days (approximately six months) and define the systemic event as a 40% market drop over this horizon. Since the model is specified in decimal log returns, the crash threshold is

$$c \equiv \log(1 - 0.40) = \log(0.60). \quad (3.13)$$

LRMES is then defined as the expected fractional equity loss over the horizon conditional on this market crash:

$$\text{LRMES}_{i,t} \equiv 1 - \mathbb{E}_t \left[\exp(R_{i,t}^{(T)}) \mid R_{m,t}^{(T)} \leq c \right], \quad (3.14)$$

The DCC-GARCH model provides a one-day conditional covariance matrix $\mathbf{H}_t$ for daily demeaned percent log returns. To obtain an approximate T -day covariance matrix for decimal log returns, we scale by the horizon and convert from percent to decimals:

$$\Sigma_t^{(T)} \approx \frac{T}{100^2} \mathbf{H}_t. \quad (3.15)$$

This scaling is an approximation that treats current conditional second moments as representative over the crisis horizon. Write $\sigma_{m,t}^2 = \Sigma_{mm,t}^{(T)}$ for the market variance, $\sigma_{i,t}^2 = \Sigma_{ii,t}^{(T)}$ for the bank variance, and $\sigma_{im,t} = \Sigma_{im,t}^{(T)}$ for the bank–market covariance, and let $\sigma_{m,t} = \sqrt{\sigma_{m,t}^2}$.

To keep the mapping from DCC moments to tail risk transparent, we use the Gaussian working distribution that underlies the DCC-QMLE estimation and treat $(R_{i,t}^{(T)}, R_{m,t}^{(T)})$ as conditionally bivariate normal with covariance $\Sigma_t^{(T)}$. Because returns are demeaned in Section 3.1, we set the conditional mean vector to zero, so the conditional distribution is fully characterized by $\Sigma_t^{(T)}$. Under this assumption, the conditional expected gross return factor in (3.14) has the closed-form expression

$$\mathbb{E}_t \left[\exp(R_{i,t}^{(T)}) \mid R_{m,t}^{(T)} \leq c \right] = \exp\left(\frac{1}{2}\sigma_{i,t}^2\right) \frac{\Phi\left(\frac{c - \sigma_{im,t}}{\sigma_{m,t}}\right)}{\Phi\left(\frac{c}{\sigma_{m,t}}\right)}. \quad (3.16)$$

Finally, substituting (3.16) into (3.14) yields $\text{LRMES}_{i,t}$. In the implementation, the ratio of $\Phi(\cdot)$ terms is evaluated using $\log \Phi(\cdot)$ for numerical stability in the left tail.

LRMES can also be exponentially approximated, which is often done in practice (Acharya et al., 2017). As a robustness check in this research, we also approximate LRMES from the dynamic beta as

$$\text{LRMES}_{i,t} \approx 1 - \exp(\beta_{i,t} \cdot (-0.40)). \quad (3.17)$$

In the baseline specification, however, we use the DCC-based plug-in approach in (3.14)–(3.16) because it directly incorporates the time-varying conditional covariance structure estimated by the model.

3.5 Diagnostics & Robustness

This section summarizes checks that support the interpretation of the DCC–GARCH moments and the SRISK series. The goal is to rule out patterns driven mainly by persistence, scaling in SRISK, or data and measurement choices. Additional outputs are in the Appendix.

- **Time-series persistence.** Aggregate SRISK is highly persistent in levels. Appendix Table 2 reports ADF and KPSS tests for SRISK and GPR (MA30), showing near-unit-root behavior in levels and stationarity in first differences. This motivates controlling for persistence and focusing on changes when relating SRISK to geopolitical risk.
- **GPR measurement.** The baseline uses GPR (MA30) to reduce high-frequency noise. Appendix Figure 1 shows the daily GPR series to check that results are not driven by the smoothing choice.
- **SRISK construction and scaling.** SRISK scales mechanically with system size and with the capital ratio parameter k . Appendix Figures 2 and 3 report asset- and equity-normalized SRISK, and Appendix Figure 4 varies k around $k = 0.08$. Higher k raises SRISK levels, while overall time patterns are similar across choices.
- **Rolling correlation sensitivity.** Rolling correlations depend on the window length. Appendix Figure 5 reports 6/12/24-month rolling correlations between SRISK and GPR, showing why these descriptive measures should be interpreted with caution.
- **Tail mapping.** LRMES is computed using the Gaussian mapping implied by the working likelihood and is compared to the beta-based exponential approximation (equation (3.17)) as a sensitivity check.

Further checks could include residual tests for the GARCH stage, fit comparisons against simple alternatives, or the use of alternative market benchmarks. These extensions are left for future work.

Chapter 4

Data

This study combines a global measure of geopolitical risk and market and balance-sheet information for the banks designated as global systemically important banks (G-SIBs) in 2025 (Financial Stability Board, 2025). The G-SIB classification is published annually by the Financial Stability Board (FSB), following the Basel Committee on Banking Supervision (BCBS) methodology, and aims to identify banks whose distress could trigger substantial negative spillovers to the global financial system. Focusing on G-SIBs is therefore appropriate for studying systemic risk measures and their relation to geopolitical risk.

Geopolitical risk is measured using the news-based Geopolitical Risk (GPR) index of Caldara and Iacoviello (2022). The GPR index is a global common risk factor and therefore identical across all banks in the sample. The series are downloaded from the authors' website and include both the daily index and its 30-day moving average (MA30). The baseline analysis uses GPR MA30 to reduce high-frequency noise and to capture more persistent geopolitical risk conditions. The unsmoothed daily series is used in robustness checks (See Appendix Figure 1).

Bank-level inputs are provided by the data department of De Nederlandsche Bank and contain market data for the G-SIBs. One of the G-SIBs (Groupe BPCE) is not publicly traded and is therefore omitted from the sample. Another G-SIB (Agricultural Bank of China) is omitted because its equity price history is substantially shorter, leaving a balanced panel of 27 G-SIBs¹ over the main sample period (27 Oct 2006 to 6 Nov 2025). For each bank, the dataset includes daily closing equity prices $P_{i,t}$, daily market value of equity $E_{i,t}$, total liabilities $L_{i,t}$ and total assets $A_{i,t}$, all in the domestic currency. The liabilities and assets are only observed at an annual frequency. To align them, the most recent reported values of $L_{i,t}$ and $A_{i,t}$ are carried forward to each trading day until the next reporting date (Brownlees & Engle, 2017).

All bank-level variables are recorded in each bank's domestic currency. Daily percentage log

¹See Table 1 in Appendix for an overview of the full sample.

returns are computed from $P_{i,t}$. The banking-sector market return is proxied by the MSCI World Banks index in local currency (LOC), which is based on constituents' local-currency returns without translation into a single base currency. This helps avoid mechanical co-movement from exchange-rate conversion. Bank returns, the MSCI index return, and changes in the GPR index are aligned to a common set of trading days. In contrast, SRISK is reported in USD to make levels comparable across banks. The inputs entering SRISK are therefore converted to USD using the corresponding daily spot exchange rates from FRED (via `pandas_datareader`).

Figure 4.1 plots the GPR MA30 series over the main sample. Pronounced spikes coincide with major events such as the Russian invasion of Ukraine in 2022 and the Israel– Hamas conflict in 2023. The post-2020 period also appears to feature a higher average level of geopolitical risk than earlier years.

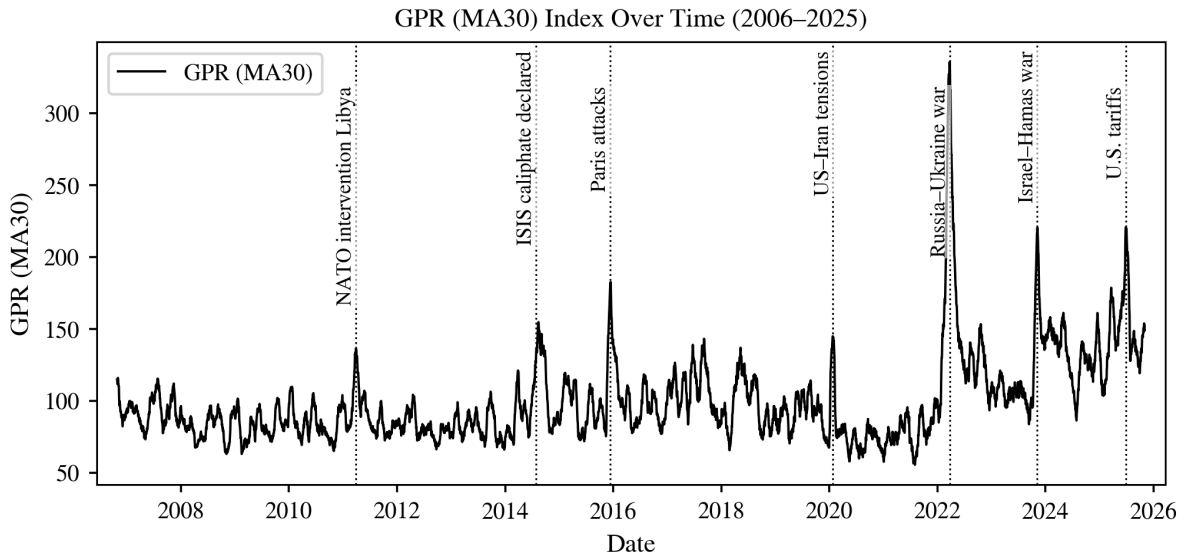


Figure 4.1: Daily 30-day moving average GPR index (Caldara & Iacoviello, 2022), 2006–2025.

In addition, Table 4.1 reports descriptive statistics for the raw daily percentage log return series. For the bank sample, statistics are computed on the pooled panel of G-SIB bank-day returns for conciseness, although patterns are similar at the individual-bank level. As is typical for daily equity returns, average returns are close to zero and the distributions exhibit skewness and excess kurtosis. The pooled minimum and maximum are relatively large because it reflects the most extreme bank-day observations in the full sample, with no outliers removed.

Table 4.1: Descriptive statistics of raw daily percentage log returns.

	Mean	Std. Dev.	Min	Median	Max	Skew	Excess Kurtosis
Pooled G-SIBs	-0.0048	2.3317	-62.5850	0.0000	89.2504	0.2821	41.6363
MSCI World Banks	-0.0042	1.3236	-12.2528	-0.0467	12.4622	0.3500	12.1505

Chapter 5

Results

This chapter presents the empirical results on the relationship between geopolitical risk and systemic risk in G-SIBs. Section 5.1 documents aggregate dynamics and evaluates whether GPR (MA30) contains predictive information for system-wide SRISK. Section 5.2 then decomposes system-wide SRISK by region to describe how contributions evolve over time.

5.1 Aggregate Dynamics

We start by looking at system-wide SRISK together with the GPR (MA30) index to give a first, aggregate view of how geopolitical risk and systemic stress move over time. Figure 5.1 plots aggregate SRISK (left axis) alongside the GPR (MA30) index (right axis) over the main sample period. Aggregate SRISK moves in broad, persistent regimes. Once it rises, it tends to stay

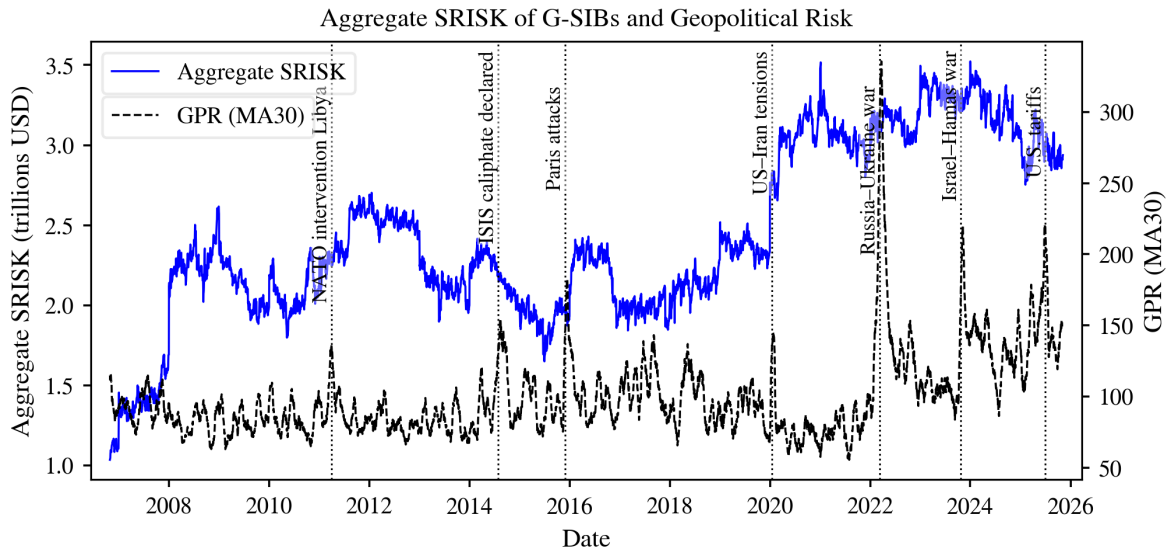


Figure 5.1: System-wide systemic risk and geopolitical risk: aggregate SRISK and GPR (MA30).

elevated for a long period and only gradually drifts down again. In the figure, SRISK increases sharply around the global financial crisis and remains high for several years, before slightly declining around the mid-2010s. SRISK increases again from 2020 onward, staying elevated through the end of the sample. The GPR series is very different. Even using the MA30 version of the index, most of the variation is concentrated in spikes around major events, after which the series quickly returns closer to a baseline level.

There are a few shared episodes where increases in geopolitical risk coincide with increases in systemic risk. One example is the coinciding spikes around the U.S.–Iran tensions in early 2020. Several high-GPR episodes occur after 2020 as well, but SRISK is also affected by non-geopolitical forces. Earlier episodes also line up with local movements in both series at times. However, the overlap does not seem systematic. Some geopolitical spikes are not clearly reflected in aggregate SRISK, and SRISK can move even when the GPR index is relatively stable. In other words, the figure suggests that geopolitical events can coincide with systemic stress, but it also shows many periods in which the two series do not move together. One reason is that the GPR index reflects different types of events, so its link to systemic stress is not consistent.

This is why the figure should be read as descriptive background rather than evidence of predictability. Because aggregate SRISK is highly persistent, correlations in levels can be driven by slow-moving components and can look different depending on the window used. Appendix Figure 5 confirms that rolling correlations vary across chosen window lengths. For that reason, the analysis below treats the co-movement as context and turns to specifications that control for persistence and focus on changes, to assess whether GPR contains any predictive power for SRISK.

It is important to note that aggregate SRISK is a level measure expressed in currency units and therefore mechanically scales with the size of the financial system and with the capital ratio parameter k used in the SRISK construction. To separate changes in scale from changes in intensity, Appendix Figures 2 and 3 report asset- and equity-normalized versions of aggregate SRISK. These scale-free series still show occasional spikes, but increases in normalized SRISK are not consistently aligned with peaks in geopolitical risk. Because SRISK is defined relative to a capital ratio k , Appendix Figure 4 shows that varying k mainly shifts the level of aggregate SRISK, without changing the broad time pattern.

Furthermore, correlation in levels can be misleading due to non-stationarity. Therefore, we assess time-series properties of aggregate SRISK and GPR (MA30) using unit-root (Dickey & Fuller, 1981) and stationarity (Kwiatkowski et al., 1992) tests. The results in Appendix Table 2 indicate that aggregate SRISK is highly persistent in levels. ADF fails to reject a unit root,

while KPSS rejects stationarity. In first differences, both tests are consistent with stationarity. For GPR, ADF rejects a unit root in levels but KPSS rejects stationarity, consistent with a mean-reverting series with episodic spikes and low-frequency shifts (Perron, 1989).

Table 5.1 evaluates whether geopolitical risk predicts aggregate SRISK at the monthly frequency. The daily aggregate SRISK and GPR (MA30) series are collapsed to month-end observations for the regression. Given the strong non-stationarity and near-unit-root persistence in SRISK levels, the regression in column (1) should not be interpreted as reliable evidence of predictability from GPR to SRISK.

Table 5.1: Predictive regressions: Monthly SRISK (trillions USD) and GPR (MA30).

	(1) SRISK_t	(2) SRISK_t (AR(1) control)	(3) ΔSRISK_t
Constant	2.4340*** (0.1047)	0.0842*** (0.0305)	0.0082 (0.0066)
GPR_{t-1}	0.1062** (0.0465)	0.0028 (0.0056)	-0.0018 (0.0056)
GPR_{t+1} (placebo)	0.1222*** (0.0443)	0.0014 (0.0054)	—
SRISK_{t-1}	—	0.9687*** (0.0122)	—
R^2	0.140	0.967	0.000
Adj. R^2	0.132	0.967	-0.004
N	228	228	228

Notes: Regressions are estimated by OLS with Newey–West (HAC) standard errors (12 lags) in parentheses. GPR is standardized to mean 0 and standard deviation 1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In column (1), lagged GPR is positively associated with SRISK. Quantitatively, a one-standard-deviation increase in GPR is associated with an approximately 106 billion USD higher month-end SRISK. However, the coefficient on the placebo lead GPR_{t+1} is also statistically significant. Given that GPR enters as a smoothed MA30 series and the data are sampled at month-end, these coefficients cannot be interpreted as genuine one-step-ahead predictability from GPR to SRISK. In column (2), controlling for SRISK persistence by including SRISK_{t-1} yields an autoregressive coefficient close to one, and the coefficient on lagged GPR becomes small and statistically insignificant. Consistent with this, column (3) shows that GPR does not predict monthly changes in SRISK. Overall, once persistence is accounted for and when focusing on changes, we find little evidence that GPR adds forecasting power for aggregate SRISK at the monthly frequency.

5.2 Regional Decomposition and Concentration

To examine cross-regional heterogeneity, G-SIBs are grouped into three different regions: North America, Europe, and Asia. This classification reflects differences in regulation, ownership structures, financial market integration, and geopolitical exposure. Regional aggregates are constructed by summing SRISK across the G-SIBs headquartered in each region. Because the sample fixes the set of institutions to the 2025 G-SIB list throughout the historical period, the decomposition should be interpreted as a consistent backcast for today’s G-SIBs rather than a time-varying membership definition. This choice may understate risk in earlier years for regions where banks later drop out of the G-SIB set.

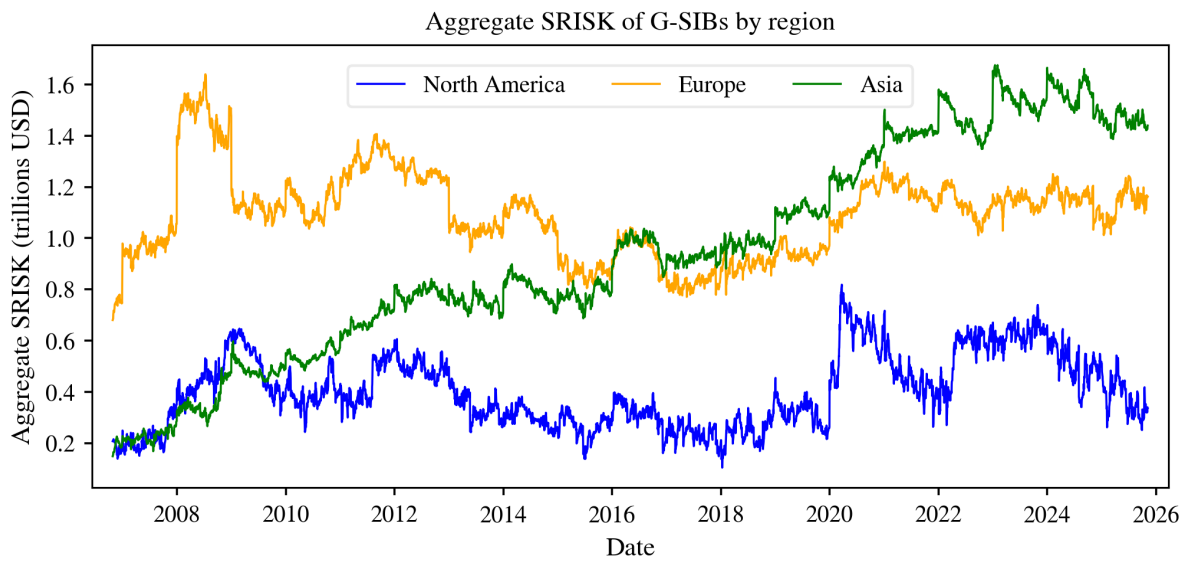


Figure 5.2: SRISK aggregated within each region (USD): North America, Europe, and Asia.

Figure 5.2 reports SRISK in levels (trillions USD) by region. Two patterns stand out. First, Europe dominates the early part of the sample, with a strong elevation around the global financial crisis and a high level in the years that follow. This likely reflects a period in which European G-SIBs are more exposed in downturns, keeping capital shortfalls high for longer. Second, Asia displays a steady increase in aggregate SRISK over the sample, eventually overtaking Europe and remaining the largest contributor in the post-2020 period. North America’s contribution is smaller for most of the sample, but it spikes sharply, especially around 2020. In other words, North America’s share can shift quickly in global stress episodes even if its contribution is typically lower.

While Figure 5.2 is informative about the absolute scale of regional systemic capital shortfalls, it is less useful for describing how regional mix changes over time. Figure 5.3 therefore reports each region’s share of total aggregate SRISK in a 100% stacked area chart, where $\text{Share}_r(t) = \sum_{i \in r} \text{SRISK}_i(t) / \sum_j \text{SRISK}_j(t)$. This is based on the same regional totals as Figure 5.2, but it makes it easier to see how the composition shifts over time. The figure shows a steady shift in the global SRISK mix away from Europe and toward Asia, while North America remains a smaller but time-varying contributor with noticeable increases during major stress episodes. This is consistent with Figure 5.2, but it makes clear that Asia’s rise reflects not only higher aggregate SRISK overall, but also a growing share of SRISK relative to Europe and North America.

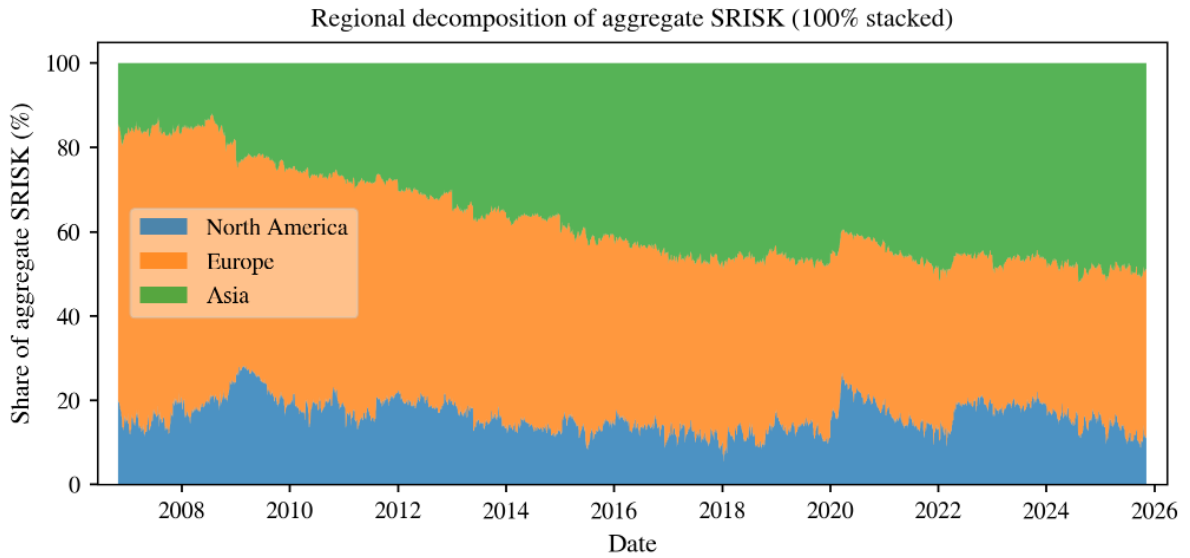


Figure 5.3: Regional shares of total aggregate SRISK (100% stacked): North America, Europe, Asia.

In line with the regional results, the rise of Asia is also visible in the concentration of SRISK among the largest institutions. In the cross-section at the end of the sample, three of the five largest SRISK contributors are Chinese banks (ICBC, Bank of China, and China Construction Bank), alongside BNP Paribas and Crédit Agricole. Holding this top-5 group fixed and tracing it back through time shows that their aggregate SRISK rises steadily, without clear crisis-like spikes, while most short-run fluctuations come from the remainder of the G-SIB set.

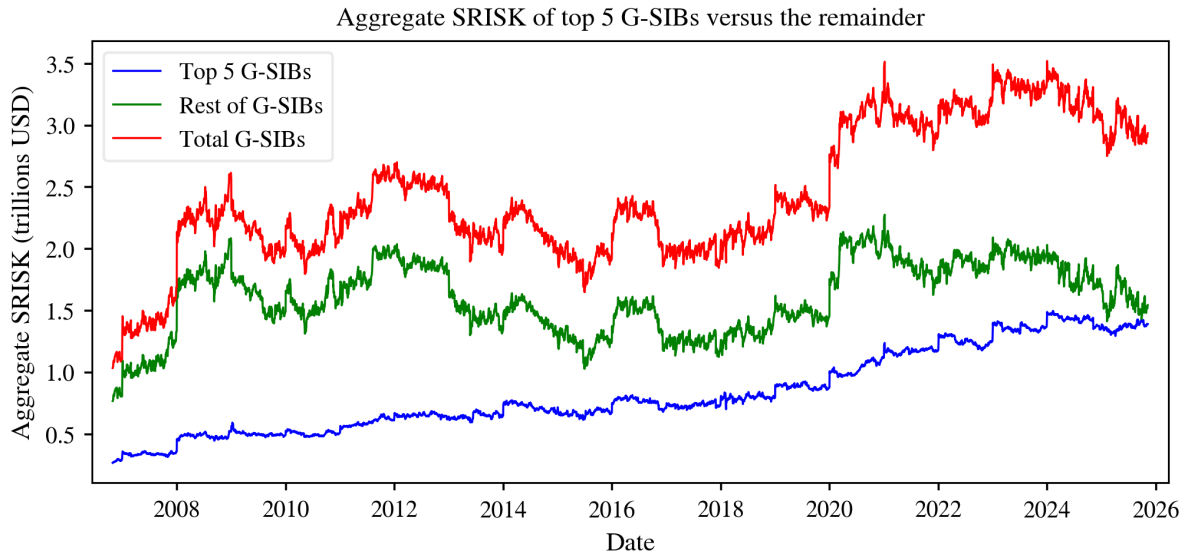


Figure 5.4: Aggregate SRISK for the top 5 G-SIBs, the remainder of G-SIBs, and the total sample.

In share terms, the top five account for just over 20% of aggregate SRISK in the late-2000s but close to 40% by the end of the sample. Because SRISK measures expected capital shortfall, large banks can contribute a lot even if their risk per unit of assets is not exceptionally high. This suggests that Asia's higher SRISK share may reflect the growing balance-sheet size of a few very large Chinese banks, rather than a broad increase in vulnerability.

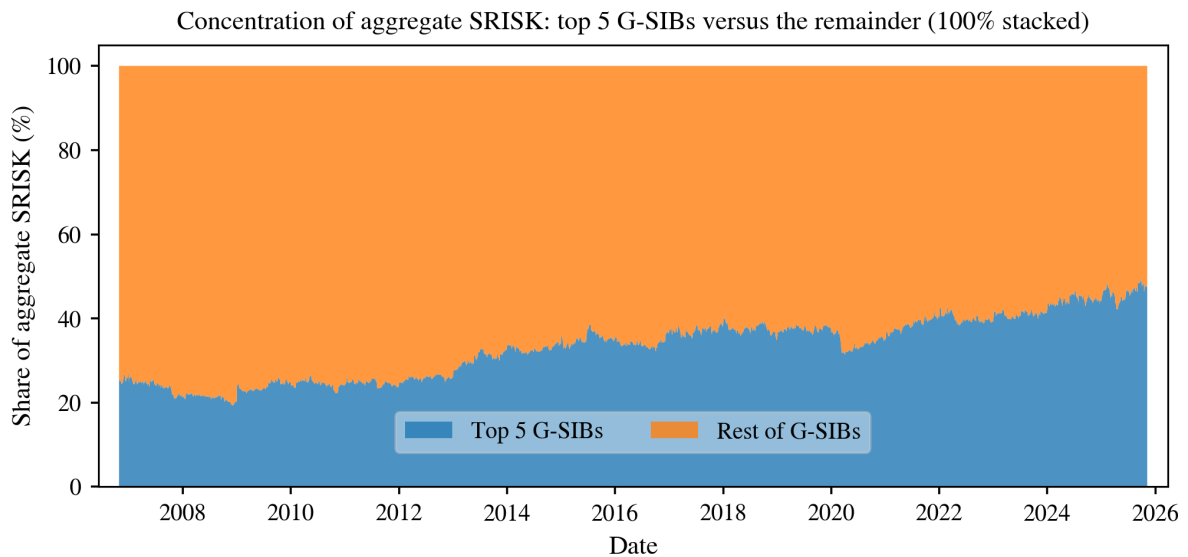


Figure 5.5: Concentration of aggregate SRISK across G-SIBs. The 100% stacked area chart reports the time-varying share of total aggregate SRISK accounted for by the top 5 G-SIBs versus the remainder of the system.

To separate size from risk, the next step normalizes aggregate SRISK by measures of bank size.

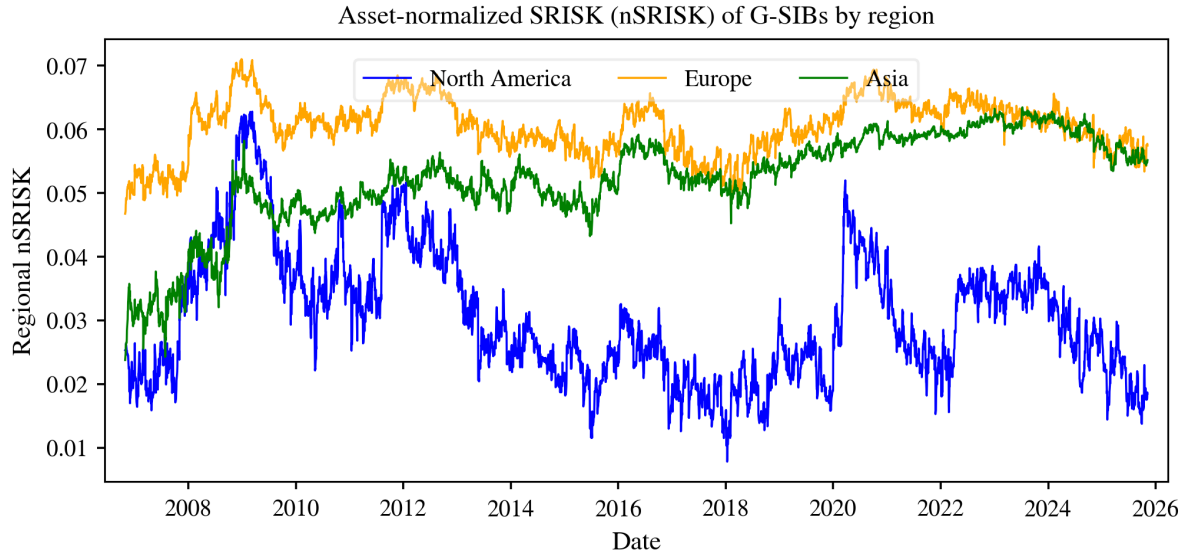


Figure 5.6: Asset-normalized SRISK (nSRISK) by region, constructed as aggregate SRISK divided by aggregate assets within each region.

Figure 5.6 reports asset-normalized regional aggregate SRISK. This measure captures the severity of potential capital shortfalls relative to balance-sheet size, and can be interpreted as the fraction of regional G-SIB assets that would need to be recapitalized in a crisis scenario. The figure shows that, once scaled by assets, Asia does not display the same strong upward trend as in aggregate SRISK levels. Instead, Asia’s normalized series remains broadly stable and converges toward Europe over time, supporting the interpretation that Asia’s growing contribution is driven mainly by size rather than a large rise in risk per unit of assets.

As a complementary view, Figure 5.7 plots the difference between each region's share of total SRISK and its share of total assets. Positive values indicate that a region contributes more SRISK than its asset share would suggest, while negative values indicate the opposite. Europe is strongly positive in the early sample, consistent with its dominant SRISK contribution around the global financial crisis. Asia moves from slightly negative to positive over time, implying that its SRISK share increasingly exceeds its asset share. This gap should be read together with Figure 5.6. Even as Asia's relative contribution rises, the asset-normalized series increases only modestly. Overall, this points to balance-sheet growth as the main driver of Asia's higher SRISK contribution, while changes in vulnerability relative to size appear limited.

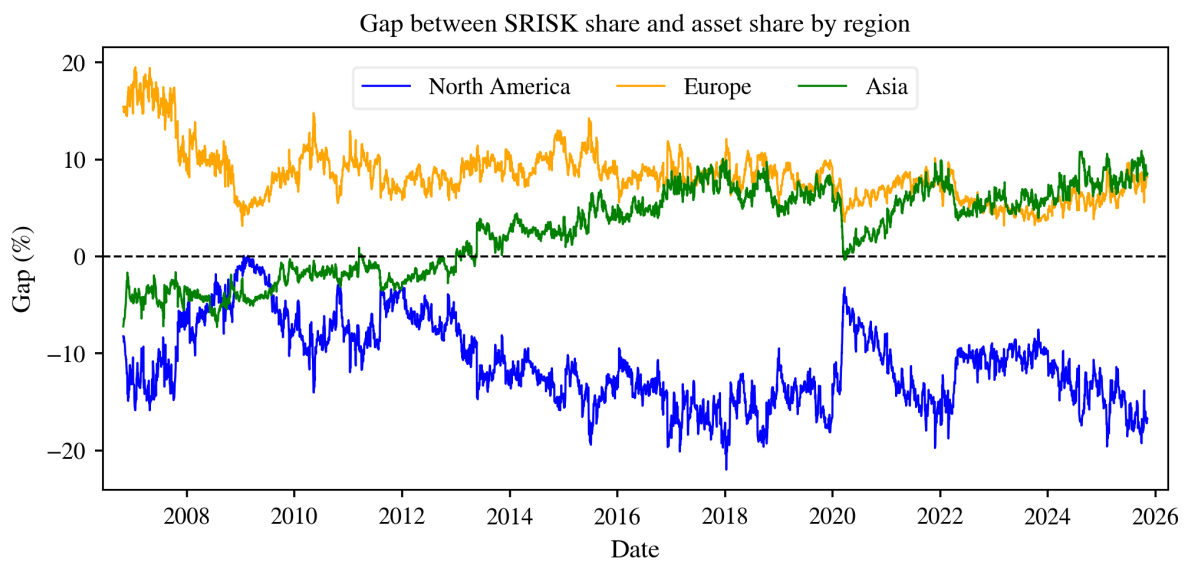


Figure 5.7: Gap between each region's share of aggregate SRISK and its share of total assets (%). Positive values indicate that SRISK share exceeds asset share.

Chapter 6

Conclusion

This thesis studied how geopolitical risk relates to systemic vulnerability in a sample of global systemically important banks (G-SIBs). Geopolitical risk was measured using the news-based GPR index of Caldara and Iacoviello (2022), and systemic vulnerability was measured using SRISK, the expected capital shortfall of banks in a severe market decline (Brownlees & Engle, 2017). The main goal was monitoring-oriented: to assess whether elevated geopolitical risk tends to coincide with higher system-wide capital shortfall risk, and to describe how the regional composition and concentration of aggregate SRISK have evolved within the 2025 G-SIB sample.

The descriptive evidence shows that spikes in geopolitical risk sometimes align with higher levels of aggregate SRISK, but the relationship is not stable over time. Aggregate SRISK tends to move in persistent patterns, whereas the GPR index is driven by occasional sharp spikes around major events. As a result, geopolitical risk does not translate automatically into higher system-wide SRISK. More importantly, once persistence is taken into account, geopolitical risk does not reliably predict aggregate SRISK at the monthly horizon. The decomposition results reveal a clear shift in where systemic vulnerability is located. In the early part of the sample, Europe accounts for the largest share of aggregate SRISK. After 2020, Asia's contribution increases significantly. At the same time, aggregate SRISK becomes more concentrated in the top-5 G-SIBs. The normalized results suggest that this mainly stems from the balance-sheet growth at a small number of very large Chinese banks. This shows that systemic risk is not only about the aggregate level, but also about where it is located and how dependent the system becomes on a few key institutions.

These findings have clear implications for macroprudential analysis and financial stability policy. The limited added predictive power of GPR suggests that geopolitical risk should be treated as a contextual indicator rather than a stand-alone warning signal for systemic capital shortfalls. At the same time, the distributional findings strengthen the case for targeted

supervision and monitoring of the largest contributors to systemic risk. Even though the analysis is mainly descriptive, it provides evidence that is directly relevant for financial stability monitoring.

Several limitations are important to keep in mind. First, the GPR index is a global measure that is identical for all banks, while geopolitical risk likely depends on bank-specific or region-specific exposures. It also behaves differently from SRISK. It is difficult to predict and moves in episodic spikes, whereas most SRISK variation is explained by its own past values. In addition, SRISK is constructed from market prices, which may already embed information about financial conditions and risk expectations. If geopolitical events are already reflected through these channels, that could help explain why the GPR index does not provide additional explanatory power. The fact that the predictability analysis uses month-end data might also mean that some short-lived effects are missed. Moreover, SRISK inherits model risk from the DCC–GARCH framework. Assumptions such as the Gaussian working distribution and the approximation used to turn the daily second moments into a six-month horizon can influence the estimated tail exposure. Finally, the sample holds the 2025 G-SIB list fixed over time. This makes the analysis a consistent backcast for today’s systemically important banks, but the results could be different in a setting where G-SIB membership varies over time.

Future research could extend this thesis in multiple directions. One natural next step would be an event-study approach around clearly dated geopolitical episodes to assess short-run responses in SRISK and its components. Another extension would be to divide geopolitical risk into more specific components, such as threats versus realized events, or to use regional measures that better reflect the exposure of different banks. Finally, estimating bank-level panel regressions that interact GPR with variables capturing different components of exposure could test whether geopolitical risk mainly affects particular business models, and whether its impact becomes stronger in periods of elevated market stress.

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Equations

Systemic Expected Shortfall

Systemic Expected Shortfall (SES) is defined as the expected value of a fraction z of bank i 's assets a^i minus its equity w^i , conditional on the event that the system's aggregate banking capital W is less than z times its aggregate assets A :

$$SES^i \equiv E [za^i - w^i | W < zA] \quad (1)$$

CoVaR

$\text{CoVaR}_q^{j|C(X^i)}(X^i)$ is implicitly defined by the $q\%$ -quantile of the conditional probability distribution:

$$\Pr(X^j | C(X^i) \leq \text{CoVaR}_q^{j|C(X^i)}) = q\%. \quad (2)$$

ΔCoVaR is defined as the additional risk to system j when institution i moves from its median state to a state of distress (its own VaR at confidence level $(1 - p)\%$). It reflects the tail dependency between the institution and the system. It is estimated using:

$$\Delta\text{CoVaR}^{j|i} = \text{CoVaR}^{j|X^i=\text{VaR}_q^i} - \text{CoVaR}^{j|X^i=\text{VaR}_{50}^i} \quad (3)$$

Tables

G-SIB sample

Table 1: G-SIB sample used in the analysis (country, institution, ISIN).

Country	Institution identified as G-SIB	ISIN
Canada	Royal Bank of Canada	CA7800871021
Canada	Toronto-Dominion Bank	CA8911605092
China	Bank of China	CNE1000001Z5
China	China Construction Bank	CNE1000002H1
China	Industrial and Commercial Bank of China	CNE1000003G1
China	Bank of Communications	CNE100000205
Germany	Deutsche Bank	DE0005140008
France	BNP Paribas	FR0000131104
France	Crédit Agricole Group	FR0000045072
France	Société Générale	FR0000130809
Japan	Mitsubishi UFJ Financial Group	JP3902900004
Japan	Mizuho Financial Group	JP3885780001
Japan	Sumitomo Mitsui Financial Group	JP3890350006
Netherlands	ING Group	NL0011821202
Spain	Banco Santander	ES0113900J37
Switzerland	UBS	CH0244767585
United Kingdom	HSBC	GB0005405286
United Kingdom	Barclays	GB0031348658
United Kingdom	Standard Chartered	GB0004082847
United States	JPMorgan Chase & Co.	US46625H1005
United States	Citigroup	US1729674242
United States	Bank of America	US0605051046
United States	Goldman Sachs	US38141G1040
United States	Bank of New York Mellon	US0640581007
United States	Morgan Stanley	US6174464486
United States	State Street	US8574771031
United States	Wells Fargo	US9497461015

Stationarity tests

Table 2: ADF and KPSS stationarity tests for aggregate SRISK and GPR MA30 (daily).

Series	Spec	ADF stat	ADF p	ADF lags	KPSS stat	KPSS p	KPSS lags	N
SRISK (level)	Const	-0.908	0.785	5	7.309	< 0.01	42	4955
SRISK (level)	Const+Trend	-2.504	0.326	5	1.044	< 0.01	42	4955
Δ SRISK	Const	-32.177	0.000	5	0.119	> 0.10	29	4954
Δ SRISK	Const+Trend	-32.188	0.000	5	0.074	> 0.10	30	4954
GPR (level)	Const	-6.013	0.000	31	3.139	< 0.01	42	4929
GPR (level)	Const+Trend	-7.060	0.000	32	0.369	< 0.01	42	4928
Δ GPR	Const	-12.565	0.000	32	0.016	> 0.10	35	4927
Δ GPR	Const+Trend	-12.567	0.000	32	0.006	> 0.10	35	4927

Notes: ADF tests the null of a unit root. KPSS tests the null of (trend-)stationarity. ADF lag length is selected by AIC. KPSS lags are selected automatically. KPSS p-values are based on lookup tables and are reported as < 0.01 or > 0.10 when the statistic falls outside the table range.

Figures

Daily GPR index

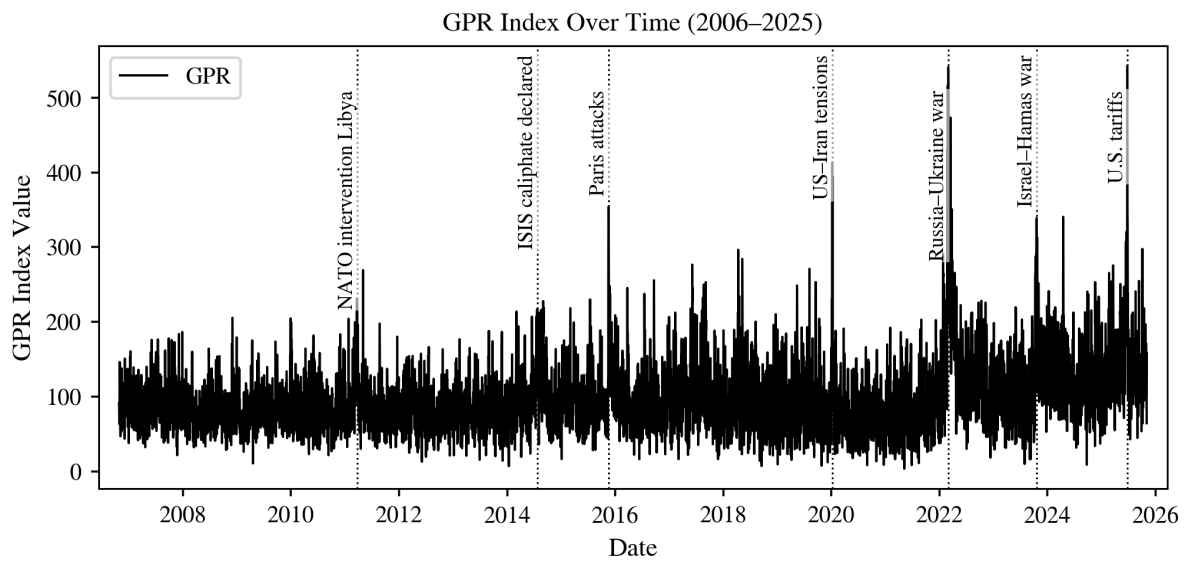


Figure 1: Geopolitical Risk (GPR) index (DAILY) with selected geopolitical events, 2006–2025.

System-wide asset-normalized SRISK (nSRISK)

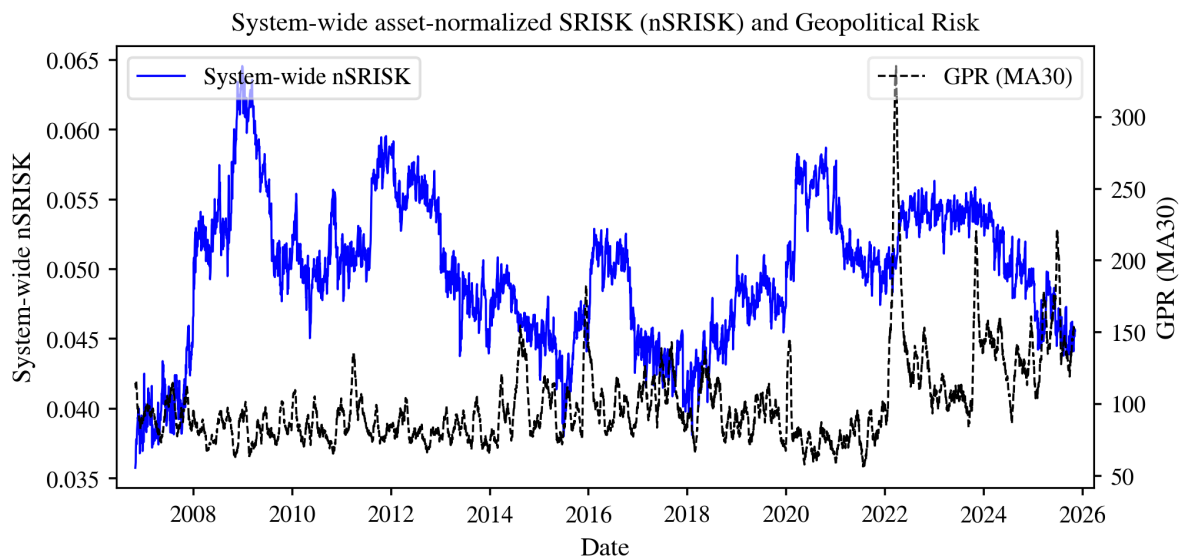


Figure 2: System-wide asset-normalized SRISK (nSRISK) for G-SIBs and the GPR index. System-wide nSRISK is defined as aggregate SRISK divided by total assets across all G-SIBs, providing a scale-free measure of systemic risk.

System-wide equity-normalized SRISK (nSRISK)

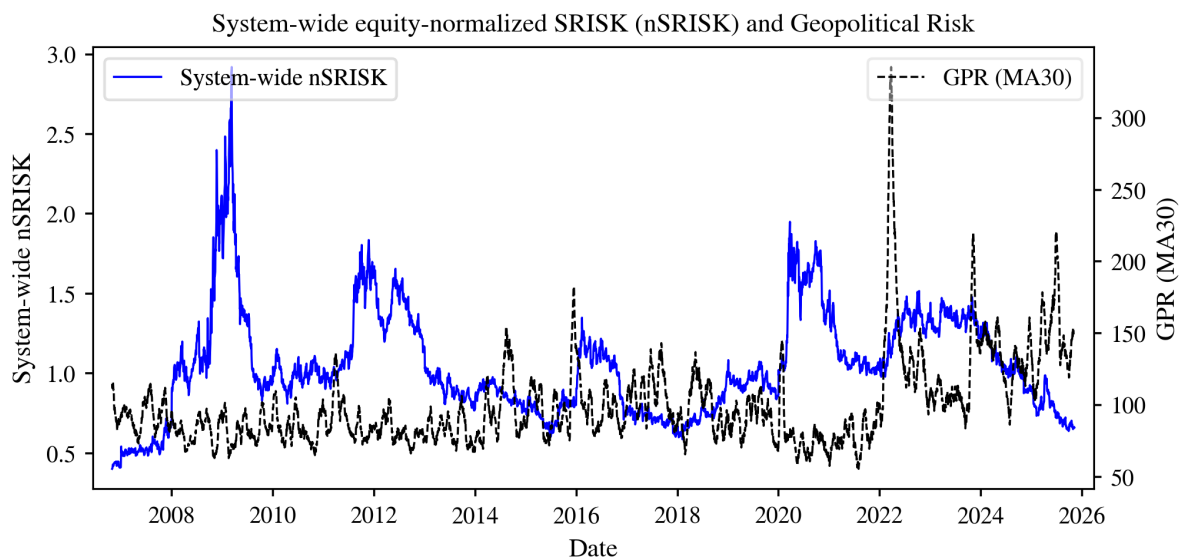


Figure 3: System-wide equity-normalized SRISK (nSRISK) for G-SIBs and the GPR index. The figure shows the evolution of aggregate SRISK relative to total system equity alongside geopolitical risk over time.

Prudential capital ratio k

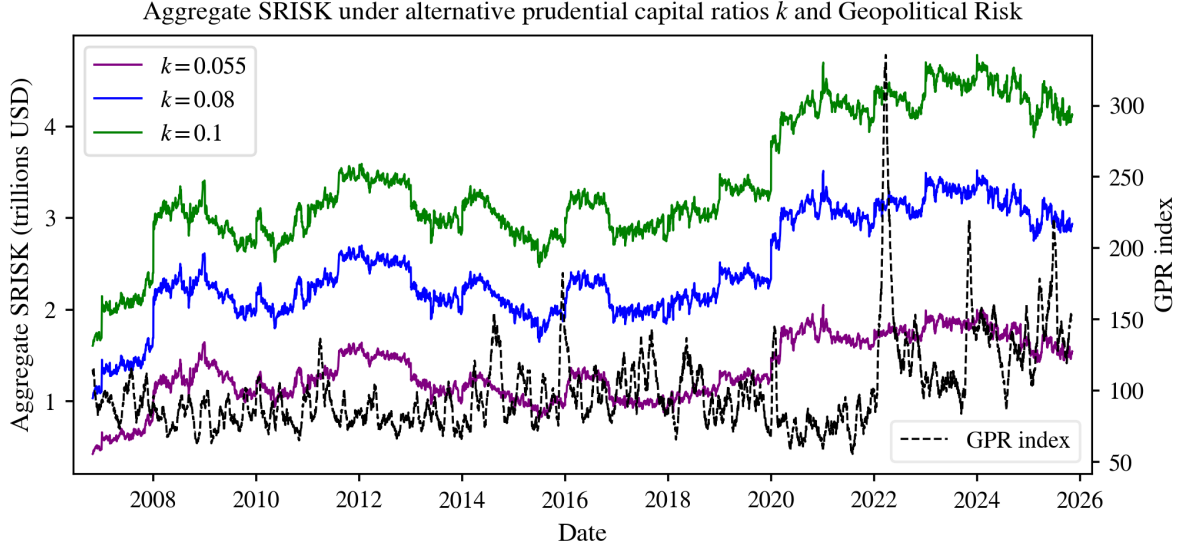


Figure 4: Aggregate SRISK (USD) for G-SIBs under alternative prudential capital ratios k , shown alongside the GPR (MA30) index. The figure illustrates how the level of aggregate SRISK varies with the choice of k over time.

For a fixed set of banks with positive shortfall, aggregate SRISK is linear in k . Differences across k therefore mainly affect the level. However, because SRISK is truncated at zero at the bank level, the set of contributing banks can change with k , so dependence measures such as correlations with GPR need not be invariant, as can be seen in Table 3.

Table 3: Correlation between aggregate SRISK and the GPR (MA30) index for alternative prudential capital ratios k .

Prudential capital ratio k	Correlation (SRISK, GPR)
0.055	0.2462
0.080	0.3178
0.100	0.3554

Notes: Given evidence of non-stationarity in levels, these correlations should not be interpreted as robust measures of dependence.

Rolling window sensitivity (6m/12m/24m)

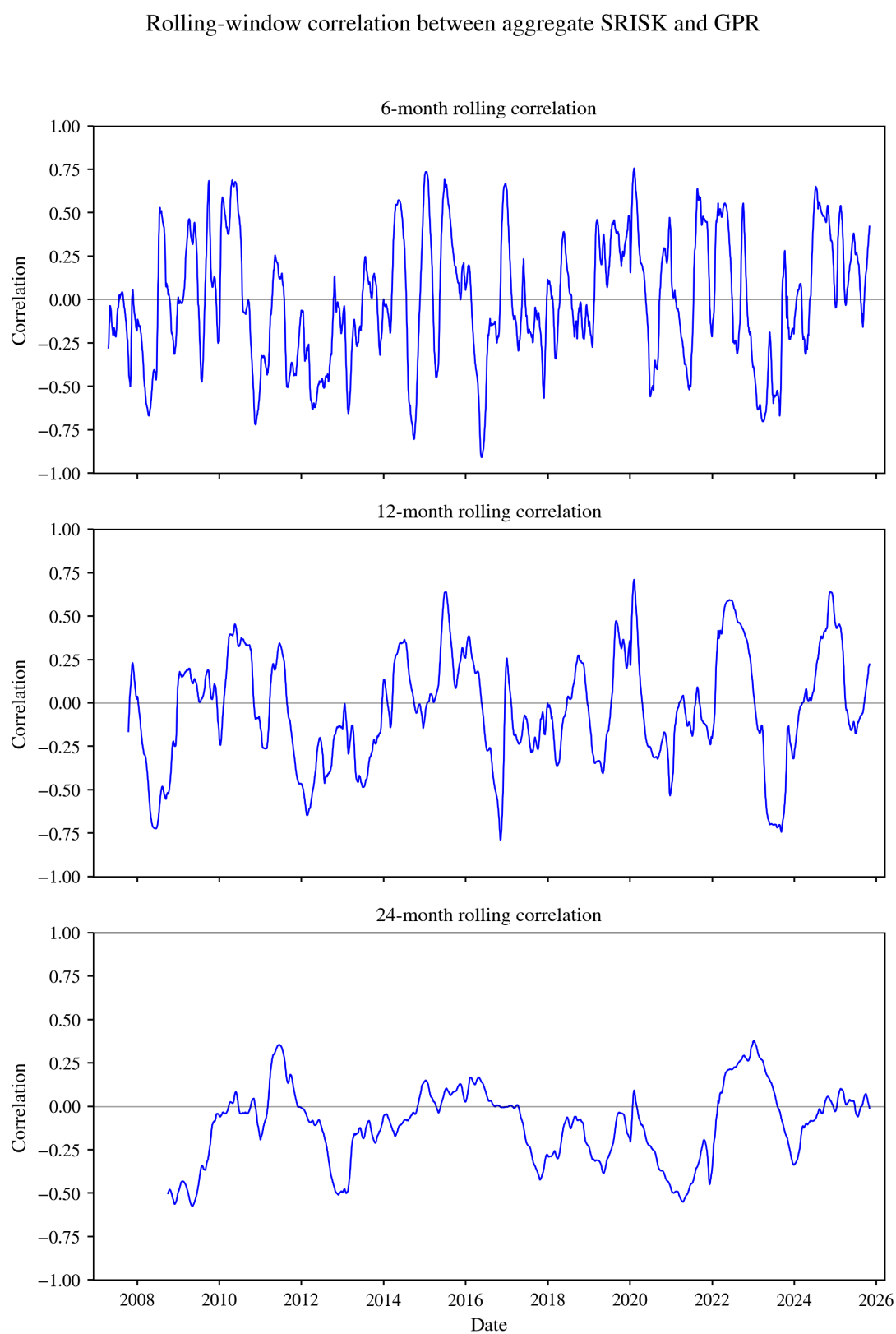


Figure 5: Rolling-window correlations between aggregate SRISK and the GPR index. Panels report correlations computed using 6-, 12-, and 24-month windows (approximately 126, 252, and 504 trading days), illustrating the sensitivity of time-varying co-movement to the choice of window length.